

# MFAD: A Multimodal Feature Fusion-Enhanced Time Series Anomaly Detection Framework in Industrial Cyber-Physical Systems

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**Abstract**—Industrial Cyber-Physical Systems (ICPS) are increasingly vulnerable to sophisticated attacks and operational disturbances that induce subtle and hard-to-detect anomalies, particularly in industrial edge environments. Existing anomaly detection methods often rely on sufficient labeled data and involve excessive computational overhead, hindering real-time detection and lightweight deployment. To address these challenges, we propose a Multimodal Feature fusion-enhanced time series Anomaly Detection framework (MFAD) in ICPS. MFAD enhances the representation of subtle anomalies by jointly modeling temporal dynamics and industrial characteristics through a unified multimodal feature fusion mechanism. Moreover, MFAD adopts a three-stage detection strategy with adaptive thresholding, which further improves robustness under varying operating conditions, while its lightweight overall architecture supports edge deployment. In addition, we provide the Industrial Gas Cyber-Physical System (IGCPS) dataset collected from real-world industrial operations. Experiments on ICPS benchmark datasets of varying scales, including IGCPS, PUMP, WADI, and SWaT, demonstrate that MFAD achieves an F1 score exceeding 96.7% with efficient resource utilization, validating its effectiveness

for real-time detection and lightweight deployment in resource-constrained industrial edge environments.

**Note to Practitioners**—This paper is motivated by the increasing need for reliable and efficient anomaly detection in Industrial Cyber-Physical Systems (ICPS), particularly deployed in resource-constrained industrial edge environments. Existing approaches often treat temporal and industrial features separately, rely on sufficient labeled data, and require substantial computational resources, which limits their applicability in real-world industrial settings. In contrast, the proposed MFAD provides a lightweight and practical solution that integrates multimodal feature fusion with robust semi-supervised detection mechanisms to effectively capture subtle anomalies in time series industrial data. The framework is designed with deployment feasibility that it offers strong detection accuracy, low latency, and efficient resource consumption suitable for industrial edge devices. The methods presented here can inform practitioners seeking to enhance the reliability and real-time performance of ICPS anomaly detection systems. Future extensions may focus on expanding MFAD for broader online industrial applications, integrating it with more edge platforms, and enabling large-scale distributed deployment.

**Index Terms**—Industrial cyber-physical systems, multimodal feature fusion, time series anomaly detection, resource-constrained industrial edge environments.

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## I. INTRODUCTION

INDUSTRIAL Cyber-Physical Systems (ICPS) constitute the operational backbone of modern industrial automation, critical infrastructure, and edge intelligence systems [1]. ICPS tightly couple computational processes with physical dynamics through interconnected sensors, actuators, controllers, and communication interfaces, thereby achieving high levels of automation, efficiency, and resilience [1], [2], [3], [4]. However, this tight cyber-physical integration also significantly enlarges the attack surface: sophisticated cyber intrusions, stealthy signal manipulations, and cross-layer disturbances may propagate across both cyber and physical domains, giving rise to subtle yet safety-critical anomalies [5], [6], [7]. In addition, the migration of ICPS toward edge-assisted architectures for real-time processing further exacerbates the challenge of deploying effective anomaly detection under stringent resource constraints [8], [9], [10].

Compared with conventional Information Technology (IT) anomaly detection tasks [11], [12], [13], anomaly detection in ICPS exhibits several distinctive challenges. First, ICPS sensor streams exhibit strong temporal dependencies and multimodal heterogeneity. Anomalies in such systems often manifest simultaneously across physical measurements and control logic [14]. Such multimodal properties hinder the effectiveness of detection methods using single features, while multimodal fusion has been extensively explored in the image domain but remains under-investigated for time series anomaly detection [15]. Second, industrial edge devices operate under strict computational, memory, and latency constraints, rendering many deep anomaly detection frameworks impractical for real-world deployment [16]. Third, severe data imbalance, incomplete labeling, and the presence of noise or distribution drift further complicate training and degrade the robustness of fully supervised models [17], [18]. As a result, designing a lightweight yet robust anomaly detection framework under edge constraints remains an open problem.

Traditional statistical and machine learning methods [19] offer interpretability and low computational cost, but their reliance on handcrafted features and large amounts of labeled data limits their ability to capture complex temporal dynamics in modern ICPS. Deep learning approaches [20], including autoencoders, recurrent networks, and attention-based models, have demonstrated strong capability in modeling temporal dependencies and nonlinear patterns. Nevertheless, most existing deep models focus on temporal features, underutilize rich industrial feature representations, and incur computational overhead that exceeds the capacity of edge devices. Moreover, the lack of effective multimodal feature fusion and adaptive thresholding mechanisms further constrains their robustness across diverse operational conditions.

To address these challenges, this paper proposes propose a Multimodal Feature fusion-enhanced time series Anomaly Detection framework (MFAD) in resource-constrained ICPS environments. MFAD constructs a unified multimodal representation by jointly modeling temporal dynamics and industrial statistical features, enabling more accurate detection of subtle cyber-physical anomalies. A lightweight architecture and semi-supervised learning paradigm are adopted to reduce computational overhead and alleviate label scarcity. Furthermore, MFAD incorporates a three-stage detection mechanism with an adaptive threshold strategy, achieving strong detection performance, robustness against Gaussian noise and distribution shifts, and consistent performance across varying industrial operating conditions.

The main contributions of this work are summarized as follows:

- 1) We propose MFAD, a lightweight Multimodal Feature fusion-enhanced time series Anomaly Detection framework that jointly models temporal dependencies and industrial domain characteristics, enabling effective cross-domain anomaly representation in ICPS environments.
- 2) We design a lightweight semi-supervised learning architecture incorporating a progressive three-stage detection mechanism with adaptive threshold calibration, which

improves robustness against Gaussian noise and distribution shifts in resource-constrained ICPS edge environments.

- 3) We contribute the Industrial Gas Cyber-Physical System (IGCPS) dataset, a real-world benchmark collected from Shenzhen Gas Corporation operational environments, containing both normal operations and cyber-physical attack scenarios.
- 4) We conduct extensive experiments on ICPS time series datasets of varying scales (IGCPS, PUMP, SWaT, and WADI), demonstrating that MFAD consistently achieves superior detection performance, with F1-scores exceeding 96.7% across all benchmarks, while maintaining efficient resource utilization and real-time detection performance for industrial edge deployment.

The remainder of this paper is organized as follows. Section II reviews existing ICPS time series anomaly detection research. Section III describes the proposed MFAD framework in detail. Section IV introduces the experimental results. Section V presents robustness evaluation and analysis. Section VI concludes the paper with future directions.

## II. RELATED WORK

This section provides a comprehensive review of anomaly detection research in industrial cyber-physical systems ICPS. Existing work can be grouped into three categories: traditional statistical and machine-learning methods, deep learning models, and advanced techniques.

### A. Traditional Methods

Traditional anomaly detection methods mainly consist of statistical modeling and classical machine learning techniques. Breunig et al. [21] introduced the Local Outlier Factor (LOF), which measures density deviation to detect local anomalies. Although LOF effectively detects contextual anomalies, its sensitivity to parameter settings limits its stability in heterogeneous ICPS environments. To model nonlinear system behaviors, Martinez-Guerra and Mata-Machuca [22] proposed fault detection schemes based on the Ensemble Kalman Filter (EnKF), which exploit recursive Bayesian estimation to identify abnormal state deviations. Although offering strong theoretical guarantees, the computational burden of EnKF increases rapidly in high-dimensional industrial processes.

Furthermore, Wu et al. [23] incorporated Bayesian learning and Gaussian processing into LSTM models for anomaly detection, enabling interpretable probabilistic reasoning over multivariate temporal data. However, the introduced Bayesian components increase inference latency and restrict real-time usage in edge devices. In parallel, Qin and Lou [24] employed Isolation Forest (IF) for hydrological anomaly detection, leveraging random partitioning to efficiently isolate rare events. Nevertheless, its lack of temporal awareness limits its applicability in dynamical ICPS environments. To support industrial streaming applications, Rosenberger et al. [25] extended Kernel Density Estimation (KDE) for real-time anomaly scoring. Despite its lightweight nature, KDE struggles with

non-stationary distributions and concept drift. Similarly, Razzak et al. [26] enhanced One-Class SVM (OCSVM) using bounded loss functions and randomized nonlinear kernels to handle large-scale IoT data. While improving robustness, the kernel operations introduce scalability issues for continuous ICPS monitoring.

Overall, traditional methods provide interpretability and computational efficiency. However, their limited capability in modeling multimodal features and complex temporal dependencies poses challenges in modern ICPS scenarios, motivating the need for data-driven sequence modeling.

### B. Deep Learning Approaches

With the success of representation learning, deep learning-based anomaly detectors have become mainstream. Park et al. [27] developed the LSTM-VAE architecture, combining recurrent networks with variational inference to model nonlinear temporal dynamics. Although effective for reconstructing multivariate sequences, this approach requires significant computation and memory. To further improve robustness, Gao et al. [28] proposed RobustTAD, integrating time-frequency decomposition with convolutional learning. Despite improved noise resilience, the heavy decomposition pipeline hinders real-time deployment. Advancing toward attention mechanisms, Yang et al. [29] introduced DCdetector, which applies dual-attention contrastive learning to enhance discriminative representations. While achieving strong performance, the dual-branch architecture incurs quadratic memory usage. Meanwhile, Su et al. [30] proposed OmniAnomaly, which incorporates planar normalizing flows into a GRU-VAE architecture to represent the stochastic characteristics of multivariate time series. Despite its robustness, the model faces challenges in handling extremely high-dimensional ICPS data.

On the one hand, Audibert et al. [31] presented USAD, a lightweight adversarial autoencoder that amplifies reconstruction discrepancies. Though training-efficient, USAD often becomes sensitive to unstable ICPS signals. On the other hand, Wang et al. [32] proposed CutAddPaste, a data augmentation strategy that creates pseudo anomalies by injecting synthetic patterns into normal sequences. Although alleviating data scarcity, these artificial patterns cannot fully capture the diversity of real industrial anomalies. To address system identification, Feng and Tian [18] developed NSIBF, a neural system identification model equipped with Bayesian filtering. Its robustness is strong, but parameter tuning is non-trivial. In addition, Kim et al. [33] proposed CTAD, which utilizes contrastive learning with multiscale augmentations to enhance noise robustness; yet, handling cross-component interactions in ICPS remains challenging. Recently, Tian et al. [34] designed SSDCL, a denoising-aware semi-supervised contrastive learning framework achieving strong anomaly tolerance, but it requires delicate hyperparameter balancing.

In summary, deep learning methods significantly improve detection accuracy and feature learning, but often introduce

high computational overhead, limiting their feasibility for real-time and resource-constrained industrial deployment.

### C. Advanced Techniques

Beyond traditional and deep learning approaches, researchers have introduced advanced generative and graph-based models to capture spatiotemporal dependencies. Han and Woo [35] introduced FuSAGNet, which integrates sparse autoencoders with graph structures, effectively modeling sensor dependencies. However, its static graph assumption limits adaptability to dynamically changing industrial relations. Similarly, Chen et al. [36] introduced DVGCRN, a deep variational graph convolutional recurrent network capable of learning spatial and temporal correlations jointly. Although effective on server and satellite data, its computational complexity restricts its applicability to edge devices. Building on graph relational learning, Zhang et al. [37] designed GRELeN, integrating GNN-based relational modeling with VAE structures. While maintaining strong anomaly discrimination, its generalization to diverse industrial scenarios remains limited. More recently, Dai and Chen [38] proposed GANF, a graph-augmented normalizing flow framework that embeds causal structure learning into invertible flows. This method provides interpretable dependency learning but suffers from high computational cost due to the invertibility constraints of flow-based models. Additionally, Lin et al. [39] introduced PddBLS-AE, an ensemble denoising autoencoder framework based on Broad Learning System (BLS) for time series anomaly detection. PddBLS-AE leverages the incremental learning capability of BLS combined with ensemble denoising strategies to achieve strong reconstruction-based anomaly detection. However, its reliance on ensemble mechanisms leads to substantial floating-point operations (FLOPs) and moderate computational overhead, limiting its applicability in resource-constrained industrial edge environments.

As systematically compared in Table I, traditional methods offer valuable interpretability and computational efficiency yet lack sophisticated multimodal fusion capabilities essential for heterogeneous ICPS data. Deep learning approaches provide superior temporal modeling and detection performance but demand significant computational resources that challenge real-time edge deployment in industrial settings. Graph-based and other emerging models capture complex relational structures and data distributions effectively but suffer from limited adaptability to dynamic conditions and substantial computational overhead. As a result, the fundamental limitation across all methodological categories is the inability to simultaneously support efficient and robust anomaly detection capabilities, lightweight computational footprint, and real-time performance under the noisy, evolving conditions characteristic of industrial environments. These identified research gaps directly motivate our proposed MFAD framework, which strategically unifies industrial feature extraction, lightweight bidirectional temporal representation learning, adaptive semi-supervised anomaly detection with dynamic thresholds to comprehensively address the stringent requirements of resource-constrained ICPS deployment.

TABLE I  
COMPARISON OF TIME SERIES ANOMALY DETECTION METHODS IN ICPS

| Type                | Method                 | Multimodal | Real-time | Lightweight | Robustness |
|---------------------|------------------------|------------|-----------|-------------|------------|
| Traditional         | LOF [21]               | ×          | ×         | ✓           | ×          |
|                     | EnKF [22]              | ×          | ×         | ×           | ✓          |
|                     | Bayesian Networks [23] | ×          | ×         | ×           | ✓          |
|                     | Isolation Forest [24]  | ×          | ✓         | ✓           | ×          |
|                     | KDE [25]               | ×          | ✓         | ✓           | ×          |
|                     | OCSVM [26]             | ×          | ×         | ✓           | ✓          |
| Deep Learning       | LSTM-VAE [27]          | ×          | ×         | ×           | ✓          |
|                     | RobustTAD [28]         | ×          | ×         | ×           | ✓          |
|                     | DCdetector [29]        | ×          | ×         | ×           | ✓          |
|                     | OmniAnomaly [30]       | ×          | ×         | ×           | ✓          |
|                     | USAD [31]              | ×          | ✓         | ✓           | ×          |
|                     | CutAddPaste [32]       | ×          | ✓         | ✓           | ×          |
|                     | NSIBF [18]             | ×          | ✓         | ✓           | ✓          |
|                     | CTAD [33]              | ×          | ✓         | ✓           | ✓          |
|                     | SSDCL [34]             | ×          | ✓         | ✓           | ✓          |
| Advanced Techniques | FuSAGNet [35]          | ✓          | ×         | ×           | ✓          |
|                     | DVGCRN [36]            | ✓          | ×         | ×           | ✓          |
|                     | GRELeN [37]            | ✓          | ×         | ×           | ✓          |
|                     | GANF [38]              | ✓          | ×         | ×           | ✓          |
|                     | PddBLS-AE [39]         | ✓          | ×         | ×           | ✓          |
| Ours                | MFAD                   | ✓          | ✓         | ✓           | ✓          |

Note: “✓” denotes strong capability; “×” denotes limited capability.

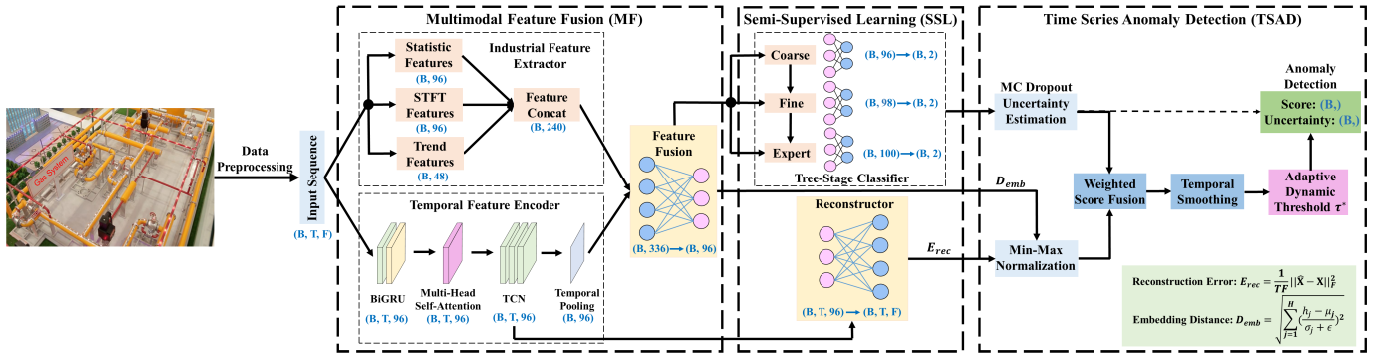


Fig. 1. Overall framework of MFAD for industrial anomaly detection.

### III. METHODOLOGY

This section provides a comprehensive description of the MFAD framework for anomaly detection in ICPS.

#### A. Overall Architecture

The MFAD model is specifically designed to extract multimodal features from the physical layer through capturing industrial characteristics and temporal dependencies of industrial processes. It also supports semi-supervised learning under limited labeled data. As shown in Fig. 1, the architecture of MFAD consists of three primary modules: (i) Multimodal Feature Fusion (MF) module, including industrial feature extraction and temporal representation learning through Bidirectional Gated Recurrent Unit–Multi-Head Self-Attention–Temporal Convolutional Network (BiGRU–Attention–TCN) architecture, (ii) Semi-Supervised Learning (SSL) module, including reconstructor and three-stage classifier, and (iii) Time Series Anomaly

Detection (TSAD) module, adaptive anomaly scoring on the output of SSL module with dynamic threshold.

Let the multivariate time series input be  $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ , where  $B$ ,  $T$ , and  $F$  denote the batch size, sequence length, and feature dimension respectively. The MFAD model outputs an anomaly score  $S \in [0, 1]$  for each sample, which indicates the probability that the input sequence exhibits abnormal behavior. The overall pipeline is formulated as:

$$S = \mathcal{F}_{\text{detect}}\left(\mathcal{F}_{\text{fusion}}\left(\mathcal{F}_{\text{temp}}(\mathbf{X}), \mathcal{F}_{\text{ind}}(\mathbf{X})\right)\right) \quad (1)$$

where  $\mathcal{F}_{\text{ind}}$  represents the industrial feature extractor,  $\mathcal{F}_{\text{temp}}$  denotes the temporal feature encoder consisting of BiGRU–Attention–TCN architecture,  $\mathcal{F}_{\text{fusion}}$  refers to the multimodal feature fusion, and  $\mathcal{F}_{\text{detect}}$  represents the final time series anomaly detection after semi-supervised learning through reconstructor and classifiers. This hierarchical design implements a three-stage information-refinement pipeline: (i) the MF module that jointly encodes industrial domain characteristics and temporal dependencies into a unified latent representation; (ii) the SSL module that performs cascaded



classification (the three-stage classifier) and reconstruction regularization to learn discriminative and robust representations; and (iii) the TSAD module that integrates MC-Dropout uncertainty estimation, weighted fusion with temporal smoothing and adaptive thresholding to produce calibrated anomaly scores. We will describe the detailed process of each module in the following sections.

### B. MF Module

MF module contains industrial feature extractor and temporal feature encoder, and fuses their representation as the foundation for the subsequent SSL module.

1) *Industrial Feature Extractor*: To capture the inherent characteristics of industrial processes, the industrial feature extractor  $\mathcal{F}_{\text{ind}}$  integrates three complementary perspectives: statistical features that capture distributional behaviors, frequency features that reflect spectral energy changes, and trend features that reveal variation dynamics. Specific feature descriptions are as follows:

a) *Statistical features*: Given the input sequence  $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ , where  $B$  denotes batch size,  $T$  denotes sequence length, and  $F$  denotes feature dimension, for each input signal, six statistics are extracted, including mean, standard deviation, minimum, maximum, median, and range:

$$\mathbf{f}_{\text{stat}} = [\mu, \sigma, \min, \max, \text{median}, \text{range}] \quad (2)$$

which describe the operational envelope and stability of each sensor. These statistics are computed along the temporal dimension, resulting in  $\mathbf{f}_{\text{stat}} \in \mathbb{R}^{B \times 6F}$ , which is then projected to  $\mathbb{R}^{B \times 96}$  through a linear layer with LayerNorm and GELU activation.

b) *Frequency features*: To identify periodic patterns or oscillations, a lightweight Short-Time Fourier Transform (STFT) with window size  $n_{\text{fft}} = 8$  and hop length = 4 is applied. The corresponding spectral power density is expressed as:

$$\mathbf{f}_{\text{freq}} = \mathbb{E}[|\mathcal{F}\{\mathbf{x}\}|^2] \quad (3)$$

which emphasizes the dominant frequencies in the signal evolution. The STFT produces  $n_{\text{freq}} = n_{\text{fft}}/2 + 1 = 5$  frequency bins per feature, yielding  $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{B \times 5F}$ , which is projected to  $\mathbb{R}^{B \times 96}$  through a dedicated frequency projection layer.

c) *Trend features*: Temporal variation is characterized by first- and second-order derivatives:

$$\mathbf{f}_{\text{trend}} = [\Delta \mathbf{x}, \Delta^2 \mathbf{x}] \quad (4)$$

reflecting both the instantaneous change rate and acceleration trend of process variables. The trend features  $\mathbf{f}_{\text{trend}} \in \mathbb{R}^{B \times 2F}$  are projected to  $\mathbb{R}^{B \times 48}$  (half the hidden dimension) to balance representation capacity.

All three components are concatenated and form the complete industrial feature representation:

$$\mathbf{z}_{\text{ind}} = \text{Concat}[\mathbf{f}_{\text{stat}}^{\text{proj}}, \mathbf{f}_{\text{freq}}^{\text{proj}}, \mathbf{f}_{\text{trend}}^{\text{proj}}] \quad (5)$$

resulting in  $\mathbf{z}_{\text{ind}} \in \mathbb{R}^{B \times 240}$  (i.e.,  $96 + 96 + 48 = 240$ ), as illustrated in the Feature Concat block of Fig. 1. This dimension is designed to match the subsequent fusion layer requirements while preserving sufficient representational capacity for industrial domain characteristics.

2) *Temporal Feature Encoder*: Temporal modeling is crucial for identifying evolving abnormal behaviors. The MFAD temporal feature encoder  $\mathcal{F}_{\text{temp}}$  integrates both bidirectional and hierarchical representations through a combination of BiGRU, Multi-Head Self-Attention, and Temporal Convolution Network (TCN).

a) *BiGRU*: Given the input sequence  $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ , an input projection layer first maps it to  $\mathbf{X}_{\text{proj}} \in \mathbb{R}^{B \times T \times H}$  where  $H = 96$  is the hidden dimension. The two-layer bidirectional GRU captures both forward and backward temporal dependencies:

$$\mathbf{h}_t = [\overrightarrow{\text{GRU}}(\mathbf{x}_{1:t}), \overleftarrow{\text{GRU}}(\mathbf{x}_{t:T})] \in \mathbb{R}^H \quad (6)$$

producing contextualized embeddings  $\mathbf{H} \in \mathbb{R}^{B \times T \times H}$ , where  $H$  is the hidden dimension of the model (number of output channels for BiGRU–Attention–TCN architecture). The BiGRU uses 2 layers with  $H/2 = 48$  hidden units per direction, and the bidirectional outputs are concatenated to maintain dimension  $H = 96$ .

b) *Multi-head self-attention*: Self-attention is employed to dynamically emphasize temporally salient events:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (7)$$

where  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$  are linear projections of  $\mathbf{H}$ , and  $d_k$  denotes the attention head dimension. This mechanism allows the model to adaptively focus on timestamps that exhibit higher anomaly potential. With 4 attention heads and per-head dimension  $d_k = H/4 = 24$ , the multi-head attention output maintains the shape  $\mathbf{H}_{\text{attn}} \in \mathbb{R}^{B \times T \times H}$  through output projection, preserving the temporal resolution for subsequent TCN processing.

c) *TCN*: The Temporal Convolutional Network (TCN) module receives the attention-enhanced sequence  $\mathbf{H}_{\text{attn}} \in \mathbb{R}^{B \times T \times H}$  and is designed to capture short-range patterns and multi-scale temporal dependencies through dilated convolutions and residual connections. Formally, for the  $l$ -th TCN layer, the output is computed as:

$$\mathbf{y}^{(l)} = \text{GELU}\left(\text{BN}(\text{Conv1D}(\mathbf{y}^{(l-1)}))\right) + \mathbf{y}^{(l-1)}, \quad (8)$$

where  $\text{Conv1D}(\cdot)$  is a one-dimensional dilated convolution,  $\text{BN}(\cdot)$  denotes batch normalization, and the non-linear activation function  $\text{GELU}(\cdot)$  is defined as:

$$\text{GELU}(x) = x \Phi(x) \quad (9)$$

with  $\Phi(\cdot)$  being the standard Gaussian cumulative distribution function (CDF). The residual connection guarantees stable gradient propagation in deep stacks of dilated convolutions. The TCN comprises 3 layers with dilation factors  $\{1, 2, 4\}$  and kernel size 3, maintaining the hidden dimension  $H = 96$  throughout via channel-preserving convolutions, yielding  $\mathbf{H}^{\text{TCN}} \in \mathbb{R}^{B \times T \times H}$ .

After the final layer, temporal aggregation is performed using global average pooling:

$$\mathbf{y}_{\text{tcn}} = \text{AvgPool}(\mathbf{y}^{(L)}) \quad (10)$$

where  $\text{AvgPool}(\cdot)$  denotes average pooling along the temporal dimension. This pooling operation compresses the temporal

dimension, producing  $\mathbf{y}_{\text{tcn}} \in \mathbb{R}^{B \times H}$  (i.e.,  $\mathbb{R}^{B \times 96}$ ), which is subsequently concatenated with the industrial features  $\mathbf{z}_{\text{ind}} \in \mathbb{R}^{B \times 240}$  to form the input to the fusion layer.

3) *Multimodal Feature Fusion*: To ensure proper temporal alignment, both the industrial feature extractor and temporal encoder operate on the same input window  $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ . Statistical features aggregate distributional properties over the complete sequence, STFT features capture frequency content through overlapping windows spanning the temporal range, and trend features compute derivatives across consecutive time steps. The temporal encoder produces sequence-level representations through average pooling after TCN processing. This synchronized computation ensures that both feature modalities represent the same temporal context before fusion. As illustrated in Fig. 1, the Feature Fusion block receives the concatenated input  $[\mathbf{y}_{\text{tcn}}; \mathbf{z}_{\text{ind}}] \in \mathbb{R}^{B \times 336}$  ( $\mathbf{y}_{\text{tcn}} \in \mathbb{R}^{B \times 96}$  and  $\mathbf{z}_{\text{ind}} \in \mathbb{R}^{B \times 240}$ ) and projects it to the target hidden dimension  $H = 96$  through a linear transformation:

$$\mathbf{h}_{\text{fusion}} = \text{GELU}(\text{LayerNorm}(\mathbf{W}[\mathbf{y}_{\text{tcn}}; \mathbf{z}_{\text{ind}}])) \quad (11)$$

where  $\text{LayerNorm}(\cdot)$  is layer normalization,  $[\cdot; \cdot]$  denotes vector concatenation, and  $\mathbf{W} \in \mathbb{R}^{96 \times 336}$  is a trainable projection matrix. This projection compresses the high-dimensional concatenated features while preserving discriminative information from both modalities, producing the unified latent representation  $\mathbf{h}_{\text{fusion}} \in \mathbb{R}^{B \times 96}$ , which serves as the shared input to both the three-stage classifier and reconstruction module. The dimensional reduction from 336 to 96 is intentionally designed to prevent overfitting while maintaining sufficient representational capacity for anomaly discrimination.

### C. SSL Module

1) *Three-Stage Cascaded Detection Strategy*: To achieve a balance among recall, discrimination robustness, and precision, the proposed framework employs a cascaded three-stage classifier with progressive refinement and adaptive early-exit.

Let  $\mathbf{h}_{\text{fusion}} \in \mathbb{R}^H$  denote the fused multimodal representation. The three-stage classifier is defined as:

$$\mathcal{F}_{\text{clf}} = \{\mathcal{F}_{\text{coarse}}, \mathcal{F}_{\text{fine}}, \mathcal{F}_{\text{expert}}\} \quad (12)$$

where each stage is a two-layer MLP with progressively higher regularization (Dropout:  $0.3 \rightarrow 0.4 \rightarrow 0.5$ ) to transition from high recall to high precision. All stages are jointly optimized with information flowing from coarse to expert through concatenation:

$$\mathbf{o}^{(1)} = \mathcal{F}_{\text{coarse}}(\mathbf{h}_{\text{fusion}}) \in \mathbb{R}^2 \quad (13)$$

$$\mathbf{o}^{(2)} = \mathcal{F}_{\text{fine}}([\mathbf{h}_{\text{fusion}}; \mathbf{o}^{(1)}]) \in \mathbb{R}^2 \quad (14)$$

$$\mathbf{o}^{(3)} = \mathcal{F}_{\text{expert}}([\mathbf{h}_{\text{fusion}}; \mathbf{o}^{(1)}; \mathbf{o}^{(2)}]) \in \mathbb{R}^2 \quad (15)$$

where  $[\cdot; \cdot]$  denotes concatenation. As shown in Fig. 1, the input dimensions progress as: Coarse stage receives  $(B, 96) \rightarrow (B, 2)$ , Fine stage receives  $(B, 98) \rightarrow (B, 2)$ , and Expert stage receives  $(B, 100) \rightarrow (B, 2)$ . The input dimensionality progressively increases from  $H$  to  $H + 2$  to  $H + 4$  implementing a cascaded refinement mechanism. Each subsequent stage receives both the original fused representation and the prediction logits from all preceding stages. This design

enables later stages to: (1) correct potential errors from earlier stages by observing their confidence distributions, (2) focus computational capacity on samples where earlier predictions exhibit uncertainty, and (3) leverage inter-stage information flow without requiring separate feature re-extraction. The 2-dimensional increments correspond to binary classification logits (normal vs. anomaly) passed between stages. Each stage outputs binary logits  $\mathbf{o}^{(s)} = [o_0^{(s)}, o_1^{(s)}]$  with classification probabilities:

$$P_k^{(s)} = \frac{\exp(o_k^{(s)})}{\sum_{j=0}^1 \exp(o_j^{(s)})}, \quad k \in \{0, 1\} \quad (16)$$

This cascaded architecture enables later stages to refine decisions by leveraging predictions from earlier stages. To reduce computational cost, samples are routed through stages based on confidence thresholds  $\tau_1 = 0.85$  and  $\tau_2 = 0.90$ . Let  $c^{(s)} = \max_k P_k^{(s)}$  denote the confidence at stage  $s$ . The routing logic is:

$$\text{Stage}(i) = \begin{cases} i = 1 & \text{if } c^{(1)} \geq \tau_1 \\ i = 2 & \text{if } c^{(1)} < \tau_1 \text{ and } c^{(2)} \geq \tau_2 \\ i = 3 & \text{if } c^{(1)} < \tau_1 \text{ and } c^{(2)} < \tau_2 \end{cases} \quad (17)$$

The final prediction uses the output from the deepest stage reached:

$$\hat{y} = \arg \max_{k \in \{0, 1\}} P_k^{(s_{\text{final}})} \quad (18)$$

where  $s_{\text{final}} \in \{1, 2, 3\}$  is determined by Eq. (17). This adaptive mechanism processes approximately 70% of samples at Stage 1, 20% at Stage 2, and 10% at Stage 3, reducing average inference cost while maintaining detection accuracy. Stage 1 (Coarse) prioritizes recall, minimizing false negatives with low dropout of 0.3, Stage 2 (Fine) balances precision and recall with moderate dropout of 0.4, and Stage 3 (Expert) enhances precision for hard samples with high dropout of 0.5. The progressive increase in input dimensionality ( $H \rightarrow H+2 \rightarrow H+4$ ) and regularization strength ensures each stage specializes in increasingly difficult discrimination tasks without deepening the network.

2) *Reconstruction Regularization*: The Reconstructor module, as shown in Fig. 1, operates on the TCN output  $\mathbf{H}^{\text{TCN}} \in \mathbb{R}^{B \times T \times 96}$  and maps it back to the original feature space through a point-wise MLP:  $\mathcal{F}_{\text{rec}} : \mathbb{R}^{96} \rightarrow \mathbb{R}^F$ , producing the reconstructed sequence  $\hat{\mathbf{X}} \in \mathbb{R}^{B \times T \times F}$ . This dimensional transformation  $(B, T, 96) \rightarrow (B, T, F)$  enables point-wise reconstruction across all timestamps for accurate anomaly localization. To preserve the smoothness and geometry of the latent normal manifold, a reconstruction branch is jointly optimized with the classifier. The reconstructor estimates an approximation of the original input sequence:

$$\hat{\mathbf{X}} = \mathcal{F}_{\text{rec}}(\mathbf{H}^{\text{TCN}}) \quad (19)$$

The reconstruction error is measured using the normalized squared Frobenius norm:

$$E_{\text{rec}} = \frac{1}{TF} \|\hat{\mathbf{X}} - \mathbf{X}\|_F^2 \quad (20)$$

which serves as a self-supervised constraint encouraging normal operational patterns to stay consistent across temporal evolutions.

3) *Joint Optimization Objective*: The total loss function integrates three-stage classification, hard sample mining, label smoothing, and reconstruction regularization:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^3 w_i \mathcal{L}_i^{\text{CE}} + w_f \mathcal{L}_{\text{focal}} + w_s \mathcal{L}_{\text{smooth}} + w_r \mathcal{L}_{\text{rec}} + w_a \mathcal{L}_{\text{anom}} \quad (21)$$

where  $\mathcal{L}_i^{\text{CE}}$  is the cross-entropy loss for stage  $i \in \{1, 2, 3\}$  (coarse/fine/expert),  $\mathcal{L}_{\text{focal}}$  is the hard example mining loss,  $\mathcal{L}_{\text{smooth}}$  is label smoothing regularization,  $\mathcal{L}_{\text{rec}}$  is normal reconstruction loss,  $\mathcal{L}_{\text{anom}}$  is anomaly reconstruction penalty, and  $(w_1, w_2, w_3, w_f, w_s, w_r, w_a) = (0.15, 0.25, 0.25, 0.20, 0.05, 0.05, 0.05)$  balance the competing objectives. To handle class imbalance [40], we use dynamic class weighting. The weighted cross-entropy loss for stage  $i$  is defined as:

$$\mathcal{L}_i^{\text{CE}} = -\frac{1}{B} \sum_{j=1}^B \omega_{y_j} \log p_i(y_j | \mathbf{x}_j) \quad (22)$$

where  $B$  is batch size,  $y_j \in \{0, 1\}$  is the ground-truth label (0 = normal, 1 = anomaly), and  $p_i(y_j | \mathbf{x}_j)$  is the predicted probability from stage  $i$ . The class weight  $\omega_{y_j}$  is determined by the sample's label according to:

$$\omega_{y_j} = \begin{cases} \omega_0 = 1.0 & \text{if } y_j = 0 \text{ (normal)} \\ \omega_1 & \text{if } y_j = 1 \text{ (anomaly)} \end{cases} \quad (23)$$

where the anomaly class weight  $\omega_1$  adapts dynamically with training progress:

$$\omega_1 = \text{clip} \left( \frac{N_0}{N_1}, 5.0, 20.0 \right) \cdot \left( 1.5 + 0.5 \cdot \frac{e}{E} \right) \quad (24)$$

where  $N_0$  and  $N_1$  respectively denote the sample counts normal and anomaly,  $e$  is the current epoch,  $E$  is total epochs, and  $\text{clip}(\cdot, a, b)$  constrains values to  $[a, b]$ . The base weight  $\frac{N_0}{N_1}$  follows the inverse frequency weighting principle [41]. The clipping bounds  $[5.0, 20.0]$  prevent numerical instability while maintaining effective class rebalancing. The progressive multiplier  $(1.5 + 0.5 \cdot \frac{e}{E})$  implements a curriculum learning schedule [42] that gradually increases emphasis on anomaly samples from  $1.5\times$  to  $2.0\times$  throughout training, allowing the model to first establish stable representations before focusing on harder anomaly discrimination tasks. To emphasize hard examples, we apply adaptive Focal Loss [41] on the expert stage:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{B} \sum_{j=1}^B (1 - p_j)^{\gamma(e)} \log(p_j + \epsilon) \quad (25)$$

where  $p_j = p_{\text{expert}}(y_j | \mathbf{x}_j)$  is clamped to  $[10^{-7}, 1 - 10^{-7}]$  for numerical stability,  $\epsilon = 10^{-8}$ , and  $\gamma(e) = 2.0 + e/E$  increases from 2.0 to 3.0. Class-aware aggregation prevents imbalanced training:

$$\mathcal{L}_{\text{focal}} = \begin{cases} 0.7 \mathcal{L}_{\text{focal}}^{\text{anom}} + 0.3 \mathcal{L}_{\text{focal}}^{\text{normal}} & \text{if } N_1 > 0 \\ \frac{1}{B} \sum_{j=1}^B \mathcal{L}_{\text{focal}}(p_j) & \text{otherwise} \end{cases} \quad (26)$$

Label smoothing with  $\epsilon = 0.05$  improves calibration [43] using fine-stage predictions:

$$\mathcal{L}_{\text{smooth}} = -\sum_k \tilde{y}_k \log p_k^{(2)}(\mathbf{x}), \quad \tilde{y}_k = \begin{cases} 0.95 & \text{if } k = y \\ 0.05 & \text{otherwise} \end{cases} \quad (27)$$

where  $k \in \{0, 1\}$  and  $p_k^{(2)}$  denotes fine-stage probability. Two reconstruction objectives enforce normal pattern learning. Given TCN output  $\mathbf{H}^{\text{TCN}} \in \mathbb{R}^{B \times T \times H}$  and reconstructor  $\mathcal{F}_{\text{rec}} : \mathbb{R}^H \rightarrow \mathbb{R}^F$ :

$$\mathcal{L}_{\text{rec}} = \frac{1}{BTF} \|\mathcal{F}_{\text{rec}}(\mathbf{H}^{\text{TCN}}) - \mathbf{X}\|_F^2$$

$$\mathcal{L}_{\text{anom}} = \begin{cases} \frac{1}{N_1 TF} \sum_{j \in \mathcal{I}_1} \|\hat{\mathbf{x}}_j - \mathbf{x}_j\|_F^2 & \text{if } N_1 > 0 \\ 0 & \text{otherwise} \end{cases} \quad (28)$$

where  $\mathcal{I}_1 = \{j : y_j = 1\}$ ,  $T$  is sequence length,  $F$  is feature dimension, and  $\|\cdot\|_F$  denotes Frobenius norm.

For unlabeled data  $\mathcal{D}_U = \{\mathbf{x}_j^u\}_{j=1}^{N_U}$  (assumed predominantly normal), we minimize reconstruction loss with weight  $w_u = 0.3$ :

$$\mathcal{L}_{\text{unsup}} = 0.3 \cdot \frac{1}{N_U TF} \sum_{j=1}^{N_U} \|\mathcal{F}_{\text{rec}}(\mathbf{H}_j^{\text{TCN}}) - \mathbf{x}_j^u\|_F^2 \quad (29)$$

Gradient clipping ( $\|\nabla\|_2 \leq 1.0$ ) and NaN handling ensure robust training:

$$\mathcal{L}_{\text{final}} = \begin{cases} \mathcal{L}_{\text{total}} & \text{if finite} \\ \mathcal{L}_2^{\text{CE}} & \text{otherwise} \end{cases} \quad (30)$$

where  $\mathcal{L}_2^{\text{CE}}$  is the fine-stage cross-entropy as a fallback.

#### D. TSAD Module

To improve detection robustness under noise and drift, the proposed framework employs a comprehensive TSAD module, as illustrated in Fig. 1. The TSAD module receives five input signals from upstream components and processes them through MC Dropout uncertainty estimation, normalization, weighted score fusion, temporal smoothing, and adaptive thresholding to produce the final anomaly detection results.

1) *Input Components*: The TSAD module integrates five complementary signals from the SSL module:

- **Classification probabilities**  $P_{\text{coarse}}, P_{\text{fine}}, P_{\text{expert}} \in \mathbb{R}^{B \times 2}$ : The anomaly probabilities from the three-stage classifier, where the second dimension represents the softmax output for the anomaly class.
- **Reconstruction error**  $E_{\text{rec}} \in \mathbb{R}^B$ : Computed from the Reconstructor as  $E_{\text{rec}} = \frac{1}{TF} \|\hat{\mathbf{X}} - \mathbf{X}\|_F^2$ , measuring the deviation between the input sequence and its reconstruction.
- **Embedding distance**  $D_{\text{emb}} \in \mathbb{R}^B$ : Derived from the fused representation  $\mathbf{h}_{\text{fusion}} \in \mathbb{R}^{B \times H}$ , quantifying the deviation from the normal manifold. The embedding distance is computed as:

$$D_{\text{emb}} = \sqrt{\sum_{j=1}^H \left( \frac{h_j - \mu_j}{\sigma_j + \epsilon} \right)^2} \quad (31)$$

where  $h_j$  is the  $j$ -th dimension of the fused embedding  $\mathbf{h}_{\text{fusion}}$ ,  $\mu_j$  and  $\sigma_j$  are the mean and standard deviation of the  $j$ -th dimension estimated from normal training samples, and  $\epsilon = 10^{-6}$  ensures numerical stability. This standardized Euclidean distance effectively measures how far each sample deviates from the learned normal manifold in the latent space.

2) *MC Dropout Uncertainty Estimation*: To quantify prediction uncertainty, the classification probabilities undergo MC Dropout sampling during inference. Dropout layers remain active at test time, and  $M = 5$  stochastic forward passes are performed. Let  $\{S^{(m)}\}_{m=1}^M$  denote the anomaly scores from these passes, the ensemble score and epistemic uncertainty are computed as:

$$\bar{S} = \frac{1}{M} \sum_{m=1}^M S^{(m)}, \quad U = \left( \frac{1}{M} \sum_{m=1}^M (S^{(m)} - \bar{S})^2 \right)^{1/2} \quad (32)$$

The uncertainty  $U \in \mathbb{R}^B$  serves three critical purposes in industrial deployment: (1) identifying fuzzy samples near decision boundaries; (2) detecting distribution shift, where systematic uncertainty increases may indicate operational regime changes or sensor drift; and (3) supporting risk-aware decision-making by enabling operators to adjust detection sensitivity based on uncertainty levels. As illustrated in Fig. 1, the uncertainty is directly output alongside the anomaly score for downstream applications.

3) *Normalization*: The reconstruction error and embedding distance are normalized to  $[0, 1]$  range using min-max normalization for consistent fusion with probability-based scores:

$$\hat{E}_{\text{rec}} = \frac{E_{\text{rec}} - E_{\text{min}}}{E_{\text{max}} - E_{\text{min}}}, \quad \hat{D}_{\text{emb}} = \frac{D_{\text{emb}} - D_{\text{min}}}{D_{\text{max}} - D_{\text{min}}} \quad (33)$$

where the normalization bounds are estimated from the training set statistics.

4) *Weighted Score Fusion*: The score fusion adapts to the stage reached by each sample during cascaded inference. Let  $s_{\text{final}} \in \{1, 2, 3\}$  denote the terminal stage determined by Eq. (17). The anomaly score is computed as:

$$S = \sum_{i=1}^{s_{\text{final}}} \alpha_i P_{\text{anom}}^{(i)} + \alpha_4 \hat{E}_{\text{rec}} + \alpha_5 \hat{D}_{\text{emb}} \quad (34)$$

where  $P_{\text{anom}}^{(i)}$  denotes the anomaly probability from stage  $i$ , and the stage-adaptive weights are defined as:

$$(\alpha_1, \alpha_2, \alpha_3) = \begin{cases} (0.85, 0, 0) & \text{if } s_{\text{final}} = 1 \\ (0.30, 0.55, 0) & \text{if } s_{\text{final}} = 2 \\ (0.20, 0.35, 0.30) & \text{if } s_{\text{final}} = 3 \end{cases} \quad (35)$$

with  $(\alpha_4, \alpha_5) = (0.10, 0.05)$  for reconstruction and embedding components. This formulation ensures that samples exiting at Stage 1 (approximately 70% of test samples) utilize only coarse-stage predictions, while harder samples benefit from refined multi-stage fusion. The stage-adaptive design reduces unnecessary computation while maintaining detection accuracy.

5) *Temporal Smoothing*: Considering that cyber-physical attacks typically persist across multiple consecutive time steps, the fused scores undergo temporal smoothing with window size  $W$ :

$$\tilde{S}_t = \frac{1}{W} \sum_{k=-\lfloor W/2 \rfloor}^{\lfloor W/2 \rfloor} S_{t+k} \quad (36)$$

The final score combines smoothed and raw components:

$$S_{\text{final}} = \lambda \tilde{S} + (1 - \lambda) \bar{S} \quad (37)$$

where  $\lambda \in (0, 1)$  controls the balance between temporal consistency and instantaneous sensitivity. This formulation captures sustained attack patterns while preserving responsiveness to sudden anomalies.

6) *Adaptive Dynamic Threshold*: The detection threshold is determined through a hierarchical search strategy that prioritizes high precision while maximizing overall detection performance. The optimal threshold  $\tau^*$  is obtained by solving:

$$\tau^* = \arg \max_{\tau} F_1(\tau) \quad \text{s.t.} \quad \text{Precision}(\tau) \geq \gamma \quad (38)$$

where  $F_1(\tau)$  is the harmonic mean of precision and recall:

$$F_1(\tau) = 2 \cdot \frac{\text{Precision}(\tau) \cdot \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)} \quad (39)$$

with:

$$\text{Precision}(\tau) = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall}(\tau) = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (40)$$

Here, TP, FP, and FN denote true positives, false positives, and false negatives at threshold  $\tau$ , respectively. The constraint parameter  $\gamma$  starts at 0.95 and relaxes to 0.90 if no feasible solution exists. The search range is  $[\mu_0 + 0.2(\mu_1 - \mu_0), \mu_1]$ , where  $\mu_0$  and  $\mu_1$  are the mean scores of normal and anomalous samples. This strategy ensures  $\text{Precision} \geq 95\%$  when achievable, minimizing false alarms in safety-critical applications. Unlike methods that only minimize  $|\text{Precision}(\tau) - 0.95|$ , our approach explicitly balances precision and recall through the F1 objective. This balance is essential for ICPS, where both false alarms and missed detections have serious operational consequences.

7) *Anomaly Detection*: The TSAD module produces two outputs with dimensions  $(B, )$ : (1) the anomaly score  $S_{\text{final}} \in \mathbb{R}^B$  computed by comparing with threshold  $\tau^*$ , yielding a binary decision per sample, and (2) the uncertainty score  $U \in \mathbb{R}^B$  quantifying prediction confidence for each sample, as illustrated in Fig. 1. The final Score and Uncertainty both have shape  $(B, )$ , representing per-sample scalar values. High uncertainty samples can be flagged for manual review, enhancing the practical utility of MFAD in industrial operations. Algorithm 1 summarizes the complete anomaly detection workflow. The procedure alternates between multimodal feature extraction, temporal encoding, fusion, hierarchical classification, and reconstruction. Both labeled and unlabeled samples are incorporated through a semi-supervised optimization loop.

Algorithm 1 summarizes the MFAD for time series anomaly detection procedure in ICPS. For each epoch, the model processes labeled mini-batches by extracting industrial features  $\mathbf{z}_{\text{ind}}$  via  $\mathcal{F}_{\text{ind}}$ , encoding temporal patterns  $\mathbf{H}^{\text{TCN}}$  via  $\mathcal{F}_{\text{temp}}$ ,



**Algorithm 1** MFAD for Time Series Anomaly Detection in ICPS

**Require:** Labeled dataset  $\mathcal{D}_L = \{(\mathbf{X}_i, y_i)\}_{i=1}^{N_L}$ , unlabeled dataset  $\mathcal{D}_U = \{\mathbf{X}_j\}_{j=1}^{N_U}$ , learning rate  $\eta$ , total epochs  $E$ , confidence thresholds  $\tau_1, \tau_2$

**Ensure:** Trained model parameters  $\theta$ , calibrated temperature  $T$

- 1: **for**  $e = 1$  **to**  $E$  **do**
- 2:   **for** each mini-batch  $(\mathbf{X}_b, \mathbf{y}_b) \in \mathcal{D}_L$  **do**
- 3:      $\mathbf{z}_{\text{ind}} \leftarrow \mathcal{F}_{\text{ind}}(\mathbf{X}_b)$  {Industrial features}
- 4:      $\mathbf{H}^{\text{TCN}} \leftarrow \mathcal{F}_{\text{temp}}(\mathbf{X}_b)$  {Temporal encoding}
- 5:      $\mathbf{h} \leftarrow \mathcal{F}_{\text{fusion}}(\mathbf{H}^{\text{TCN}}, \mathbf{z}_{\text{ind}})$  {Feature fusion}
- 6:      $\{\mathbf{o}^{(s)}\}_{s=1}^3 \leftarrow \mathcal{F}_{\text{clf}}(\mathbf{h})$  {Three-stage classification}
- 7:      $\hat{\mathbf{X}} \leftarrow \mathcal{F}_{\text{rec}}(\mathbf{H}^{\text{TCN}})$  {Reconstruction}
- 8:     Compute  $\mathcal{L}_{\text{total}}$  via Eq. (21)
- 9:      $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{total}}$
- 10:   **end for**
- 11:   **for** each mini-batch  $\mathbf{X}_u \in \mathcal{D}_U$  **do**
- 12:     Compute  $\mathcal{L}_{\text{unsup}}$  and update  $\theta$
- 13:   **end for**
- 14: **end for**
- 15: Calibrate temperature  $T$  on validation set
- 16: Compute adaptive threshold  $\tau^*$  via precision-constrained search

and fusing both modalities through  $\mathcal{F}_{\text{fusion}}$  to obtain the unified representation  $\mathbf{h}$ . The three-stage classifier  $\mathcal{F}_{\text{clf}}$  produces hierarchical predictions, while the reconstruction module  $\mathcal{F}_{\text{rec}}$  generates  $\hat{\mathbf{X}}$  for auxiliary supervision. The total loss  $\mathcal{L}_{\text{total}}$  (Eq. (21)) is computed and model parameters  $\theta$  are updated via gradient descent:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}_{\text{total}} \quad (41)$$

where  $\eta$  denotes the learning rate controlling optimization step size. For unlabeled data, the model computes unsupervised losses  $\mathcal{L}_{\text{unsup}}$  to leverage unlabeled samples in semi-supervised learning. After training, temperature calibration adjusts probabilistic confidence on a validation set, and the adaptive threshold  $\tau^*$  is determined via precision-constrained search to optimize anomaly detection decisions.

#### E. Computational Complexity and Deployment Analysis

To evaluate the practicality of MFAD for real-world industrial scenarios, we analyze both computational efficiency and edge deployment feasibility.

Let  $T$  denote the sequence length,  $F$  the feature dimension,  $H$  the hidden dimension, and  $K$  the number of attention heads. The total computational cost of MFAD can be expressed as:

$$\mathcal{O}_{\text{MFAD}} = \mathcal{O}(TFH) + \mathcal{O}(TH^2) + \mathcal{O}(HKd) \quad (42)$$

where the first term corresponds to the BiGRU encoder, the second to the multi-head self-attention with head dimension  $d = H/K$ , and the last term to the TCN convolutional layers.

TABLE II  
HYPERPARAMETER CONFIGURATION

| Hyperparameter                       | Value       | Search Range                           |
|--------------------------------------|-------------|----------------------------------------|
| Hidden dimension $H$                 | 96          | {64, 96, 128}                          |
| Learning rate $\eta$                 | $8e^{-4}$   | $\{e^{-4}, 5e^{-4}, 8e^{-4}, e^{-3}\}$ |
| Dropout (Coarse/Fine/Expert)         | 0.3/0.4/0.5 | {0.2-0.5} step 0.1                     |
| Batch size                           | 64          | {32, 64, 128}                          |
| Confidence threshold $\tau_1/\tau_2$ | 0.85/0.90   | {0.80-0.95}                            |
| Weight decay                         | $5e^{-4}$   | $\{e^{-4}, 5e^{-4}, e^{-3}\}$          |
| MC samples                           | 5           | {3, 5, 7}                              |
| Smoothing window $W$                 | 7           | {3, 5, 7, 9}                           |
| Smoothing weight $\lambda$           | 0.6         | {0.5, 0.6, 0.7, 0.8}                   |

#### IV. EXPERIMENTS

This section introduces the evaluation metrics, the Industrial Gas Cyber-Physical System (IGCPS) dataset and experimental results for the ICPS datasets, respectively.

##### A. Experimental Environments and Hyperparameter Settings

To better approximate practical industrial edge-device conditions, all experimental simulations were conducted on an NVIDIA Jetson Nano B01 platform equipped with an ARMv8-based processor and 4 GB RAM. The trained MFAD model was exported to ONNX format and deployed using NVIDIA TensorRT with FP16 precision optimization, enabling hardware-aware inference acceleration on the embedded GPU. This setup provides a resource-constrained yet realistic computational environment for evaluating the deployment feasibility, runtime performance, and hardware-adaptive efficiency of edge-oriented anomaly detection models.

To ensure robust performance and fair comparison, all hyperparameters are selected via empirical validation and grid-based exploration on a held-out validation set, as summarized in Table II. The hidden dimension is fixed to  $H = 96$ , providing a favorable trade-off between representation capacity and computational efficiency under edge deployment constraints, while the learning rate is set to  $\eta = 8e^{-4}$ , chosen from  $\{e^{-4}, 5e^{-4}, 8e^{-4}, e^{-3}\}$  based on convergence stability and validation performance. To mitigate overfitting under class imbalance and semi-supervised learning, a progressive dropout strategy is adopted in the three-stage classifier, with dropout rates of 0.3, 0.4, and 0.5 for the Coarse, Fine, and Expert stages, respectively. During inference, confidence thresholds  $\tau_1 = 0.85$  and  $\tau_2 = 0.90$  are used to enable early-exit decisions based on prediction confidence, allowing a large proportion of easy samples to terminate at earlier stages in practice and thus reducing average inference latency. The batch size is set to 64, weight decay to  $5e^{-4}$ , and Monte Carlo dropout is performed with 5 stochastic forward passes to balance uncertainty estimation quality and efficiency. For temporal smoothing, the window size  $W = 7$  is selected based on two considerations: (1) odd-sized windows enable symmetric centered smoothing, and (2) empirical analysis indicates that most ICPS attacks persist for 5-10 consecutive time steps. The smoothing weight  $\lambda = 0.6$  is determined via grid search to balance temporal consistency with instantaneous sensitivity.

The final configuration is selected according to the highest validation F1-score.

### B. Baselines and Evaluation Metrics

To provide a fair comparison, we benchmark our method against recent representative anomaly detection algorithms in ICPS, including LSTM-VAE (2018) [27], USAD (2020) [31], NSIBF (2021) [18], FuSAGNet (2022) [35], CTAD (2023) [14], DCdetector (2023) [15], and SSDCL (2025) [16], PddBLS-AE (2025) [39], and we reproduced their work based on relevant papers. The evaluation follows a standard binary classification protocol in which each test sample is assigned to one of four outcome categories: TP — the model correctly labels an anomalous sequence as abnormal; FN — an abnormal sequence is mistakenly classified as normal; TN — a normal sequence is correctly recognized as normal; FP — a normal sequence is incorrectly flagged as anomalous.

To quantify detection performance, we employ three widely used metrics: Precision (Pre) (Eq.(40)), Recall (Rec) (Eq.(40)), and F1-Score (F1) (Eq.(39)). Precision measures how many of the predicted anomalies are truly abnormal, while Recall reflects the proportion of actual anomalies successfully detected. The F1-Score represents their harmonic mean, balancing both detection accuracy and coverage.

Beyond detection quality, we also assess the computational efficiency of the proposed framework using three additional indicators: inference latency per sample (Infer), CPU usage (CPU), and FLOPs, which collectively capture the suitability of our method for deployment in resource-constrained industrial edge environments.

### C. ICPS Time Series Datasets

To thoroughly assess the performance of the MFAD framework, we carried out an extensive experimental study on multiple ICPS time series datasets. The evaluation covered our internally collected IGCPs dataset in addition to three widely used public benchmarks: PUMP, WADI, and SWaT.

We split all datasets into a training set, a validation set, and a test set following a semi-supervised learning paradigm. The training set contains both labeled and unlabeled data: 5% labeled samples are used for supervised classification training, while the remaining 95% unlabeled samples (assumed predominantly normal operational data) are leveraged for self-supervised reconstruction learning to capture normal behavioral patterns. The validation set is used for hyperparameter tuning, temperature calibration, and early stopping. The test set is used for final model evaluation and performance validation.

1) *IGCPs Dataset*: The IGCPs dataset was obtained from the real operational environment of Shenzhen Gas Corporation (Guangdong, China), where both cyber-layer network traffic and physical-layer process variables were continuously recorded. The data capture includes normal operating conditions as well as intentionally induced malfunction states triggered by cyber-physical attack simulations.



Fig. 2. Experimental edge-computing platform: NVIDIA Jetson Nano B01 with TensorRT-accelerated deployment.

TABLE III  
DESCRIPTIONS OF PHYSICAL-LAYER FEATURES IN THE IGCPs DATASET

| Feature ID | Description                                    |
|------------|------------------------------------------------|
| 1          | High-pressure switch state                     |
| 2          | High-pressure sensor inlet pressure            |
| 3          | High-pressure skid flow sensor reading         |
| 4          | High-pressure sensor outlet pressure           |
| 5          | Sub-high-pressure switch state                 |
| 6          | Medium-pressure skid heater switch state       |
| 7          | Medium-pressure skid heater inlet temperature  |
| 8          | Sub-high-pressure skid flow measurement        |
| 9          | Sub-high-pressure skid outlet pressure         |
| 10         | Sub-high-pressure skid outlet temperature      |
| 11         | Medium-pressure skid flow rate                 |
| 12         | Medium-pressure skid heater outlet temperature |
| 13         | Medium-pressure skid outlet temperature        |
| 14         | Medium-pressure skid outlet pressure           |

Two representative adversarial behaviors were introduced to assess anomaly detection performance:

- **Shutdown-type attacks**: Malicious control commands or forged packets are injected to interrupt key gas supply procedures, forcing the closure of valves or disabling compressors. These actions directly compromise system availability and cause abrupt operational stops.
- **Parameter manipulation attacks**: Attacks of this category subtly alter regulatory parameters, such as pressure limits or flow rate setpoints, with the intention of misleading the control logic. Unlike shutdown attacks, the resulting anomalies evolve gradually, making them considerably more challenging to detect in real time.

All experiments were conducted under controlled conditions on April 24, 2025, between 15:17:08 and 15:31:59. During this time window, physical-layer signals exhibiting deviations in pressure, temperature, and flow rate were collected. The resulting dataset contains 895 time stamped samples with 14 monitored sensor attributes. Fig. 3 illustrates the statistical

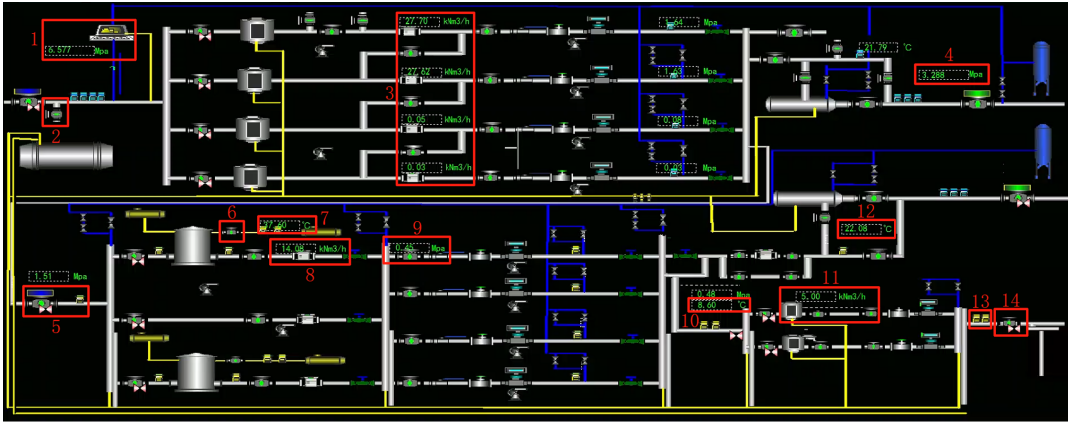


Fig. 3. Physical layer characteristics distribution and simulation scenarios in IGCPS.

distribution of these physical-layer variables, and the meaning of each feature is summarized in Table III.

In addition to the above physical indicators, the dataset also includes controller status, controller command values, and overall operational state variables. The final dataset contains 542 normal instances and 353 anomalous instances.

2) *Public Benchmarks*: Our experimental evaluation encompasses three real-world industrial Cyber-Physical System datasets: PUMP, WADI, and SWaT, representing diverse operational scenarios and scale characteristics. The fundamental attributes and distinctive features of these benchmark datasets are elaborated as follows:

The PUMP dataset originates from a municipal water treatment and distribution facility, where 52 sensing devices were installed. Data were recorded at one-minute intervals across a five-month period. Each instance is labeled according to the operational condition of the pump, namely normal, recovering, or broken.

The SWaT dataset was collected from an industrial-scale water purification testbed containing 25 sensors and 26 actuators. The full trace spans 11 days, consisting of seven consecutive days of normal plant operation followed by four days during which a variety of cyber-physical attacks were executed.

The WADI dataset extends the SWaT architecture into a water distribution scenario. It includes 67 sensors and 26 actuators, with data gathered over 16 days in total: 14 days reflecting nominal behavior and 2 days incorporating attack events.

#### D. Analysis of Anomaly Detection Results

Table IV presents the comprehensive experimental results across four ICPS datasets of varying scales, comparing MFAD against eight state-of-the-art baselines. On the IGCPS dataset, MFAD achieves 98.98% precision, 95.10% recall, and 97.00% F1-score. Among baselines, PddbLS-AE attains the closest performance with 95.05% F1, followed by SSDCL at 89.64%. NSIBF and CTAD exhibit lower F1-scores of 32.14% and 66.96% respectively, primarily due to limited recall. Regarding computational efficiency, USAD demonstrates the fastest

inference at 0.23 ms with minimal CPU usage of 1.32%, while DCdetector incurs the highest overhead at 48.36 ms inference time and 235.43 M FLOPs. MFAD maintains moderate resource consumption with 0.46 ms latency and 0.99 M FLOPs.

On the PUMP dataset, MFAD attains 99.95% precision, 99.11% recall, and 99.93% F1-score. SSDCL follows with 97.55% F1, while DCdetector achieves perfect 100% recall but lower precision at 91.12%, yielding 95.31% F1. USAD shows the largest precision-recall imbalance with 96.72% precision but only 60.61% recall, resulting in 74.52% F1. NSIBF achieves the lowest FLOPs at 0.22 M but with 95.61% F1. MFAD requires 0.58 ms inference time and 1.19 M FLOPs, representing 5.4 times lower computational cost than NSIBF while achieving 4.32% higher F1-score.

On the SWaT dataset, MFAD records 99.53% precision, 99.28% recall, and 99.40% F1-score. SSDCL ranks second with 93.02% F1, exhibiting 6.38% lower performance. FuSAGNet and CTAD achieve high precision of 98.01% and 97.56% respectively, but their recall values of 73.70% and 73.15% limit F1-scores to 84.11% and 83.65%. PddbLS-AE attains 88.72% F1 with 60.37 M FLOPs, which is 50.7 times higher than MFAD. DCdetector requires 264.52 M FLOPs, the largest among all methods, while achieving only 89.44% F1.

On the challenging WADI dataset with 5.8% anomaly ratio and 67 sensors, performance degradation is observed across all baselines. MFAD achieves 98.58% precision, 94.89% recall, and 96.70% F1-score. PddbLS-AE follows with 87.25% F1, representing a 9.45% gap. DCdetector exhibits extreme imbalance with 54.49% precision and 98.48% recall, yielding 70.17% F1. SSDCL shows similar patterns with 64.02% precision and 88.82% recall. LSTM-VAE and USAD demonstrate severe recall deficiency at 19.06% and 15.72% respectively, resulting in F1-scores below 32%. NSIBF achieves the lowest FLOPs at 0.29 M but with only 33.52% F1. DCdetector requires 624.22 M FLOPs, while MFAD maintains 1.64 M FLOPs with 26.53% higher F1-score.

Fig. 4 provides a unified radar-based comparison of detection accuracy and computational efficiency across ICPS datasets of varying scales, namely IGCPS, PUMP, SWaT, and WADI. Each subfigure jointly visualizes precision, recall,

TABLE IV  
EXPERIMENTAL RESULTS OF ANOMALY DETECTION ON DIFFERENT SCALES OF ICPS DATASETS

| Dataset | Method                 | Pre (%)      | Rec (%)       | F1 (%)       | Infer (ms)  | CPU (%)     | FLOPs (M)   |
|---------|------------------------|--------------|---------------|--------------|-------------|-------------|-------------|
| IGCPS   | LSTM-VAE (2018) [27]   | 64.10        | 73.53         | 68.49        | 1.80        | 19.52       | 4.14        |
|         | USAD (2020) [31]       | 63.11        | 75.49         | 68.75        | <b>0.23</b> | <b>1.32</b> | 2.13        |
|         | NSIBF (2021) [18]      | 29.51        | 35.29         | 32.14        | 0.41        | 8.31        | <b>0.19</b> |
|         | FuSAGNet (2022) [35]   | 68.33        | 80.39         | 73.87        | 4.83        | 8.43        | 11.02       |
|         | CTAD (2023) [33]       | 61.48        | 73.53         | 66.96        | 0.19        | 18.41       | 0.41        |
|         | DCdetector (2023) [29] | 73.58        | 90.27         | 81.08        | 48.36       | 84.83       | 235.43      |
|         | SSDCL (2025) [34]      | 94.81        | 87.88         | 89.64        | 0.39        | 11.61       | 1.92        |
|         | PddBLS-AE (2025) [39]  | 96.00        | 94.12         | 95.05        | 2.85        | 32.21       | 1.09        |
|         | <b>MFAD (Ours)</b>     | <b>98.98</b> | <b>95.10</b>  | <b>97.00</b> | 0.46        | 12.32       | 0.99        |
| PUMP    | LSTM-VAE (2018) [27]   | 94.83        | 62.44         | 75.34        | 1.84        | 17.20       | 4.85        |
|         | USAD (2020) [31]       | 96.72        | 60.61         | 74.52        | <b>0.15</b> | <b>2.12</b> | 2.26        |
|         | NSIBF (2021) [18]      | 98.22        | 93.13         | 95.61        | 0.43        | 9.12        | <b>0.22</b> |
|         | FuSAGNet (2022) [35]   | 97.13        | 83.02         | 89.52        | 5.16        | 9.31        | 11.13       |
|         | CTAD (2023) [33]       | 93.14        | 94.12         | 93.63        | 0.17        | 16.91       | 0.38        |
|         | DCdetector (2023) [29] | 91.12        | <b>100.00</b> | 95.31        | 49.33       | 81.62       | 228.53      |
|         | SSDCL (2025) [34]      | 96.95        | 98.04         | 97.55        | 0.52        | 15.23       | 1.66        |
|         | PddBLS-AE (2025) [39]  | 97.82        | 95.42         | 96.12        | 2.47        | 30.52       | 1.22        |
|         | <b>MFAD (Ours)</b>     | <b>99.95</b> | 99.11         | <b>99.93</b> | 0.58        | 8.52        | 1.19        |
| SWaT    | LSTM-VAE (2018) [27]   | 75.34        | 60.65         | 73.34        | 1.91        | 18.34       | 4.96        |
|         | USAD (2020) [31]       | 95.74        | 77.94         | 85.91        | <b>0.36</b> | <b>2.42</b> | 2.39        |
|         | NSIBF (2021) [18]      | 91.55        | 88.74         | 90.12        | 0.46        | 9.62        | <b>0.25</b> |
|         | FuSAGNet (2022) [35]   | 98.01        | 73.70         | 84.11        | 5.24        | 9.82        | 11.22       |
|         | CTAD (2023) [33]       | 97.56        | 73.15         | 83.65        | 0.39        | 17.61       | 1.17        |
|         | DCdetector (2023) [29] | 94.53        | 84.87         | 89.44        | 50.38       | 79.32       | 264.52      |
|         | SSDCL (2025) [34]      | 96.82        | 89.52         | 93.02        | 0.94        | 14.82       | 3.98        |
|         | PddBLS-AE (2025) [39]  | 96.69        | 82.28         | 88.72        | 2.50        | 31.12       | 1.37        |
|         | <b>MFAD (Ours)</b>     | <b>99.53</b> | <b>99.28</b>  | <b>99.40</b> | 0.69        | 9.61        | 1.19        |
| WADI    | LSTM-VAE (2018) [27]   | 91.28        | 19.06         | 31.45        | 1.97        | 21.63       | 5.42        |
|         | USAD (2020) [31]       | 98.11        | 15.72         | 27.12        | 0.27        | <b>2.72</b> | 2.52        |
|         | NSIBF (2021) [18]      | 35.62        | 31.62         | 33.52        | 0.49        | 9.92        | <b>0.29</b> |
|         | FuSAGNet (2022) [35]   | 80.77        | 48.95         | 60.94        | 5.62        | 10.12       | 11.36       |
|         | CTAD (2023) [33]       | 59.63        | 35.94         | 44.83        | <b>0.21</b> | 19.22       | 1.61        |
|         | DCdetector (2023) [29] | 54.49        | 98.48         | 70.17        | 55.41       | 73.82       | 624.22      |
|         | SSDCL (2025) [34]      | 64.02        | 88.82         | 74.41        | 0.79        | 8.42        | 4.05        |
|         | PddBLS-AE (2025) [39]  | 95.42        | 80.36         | 87.25        | 2.53        | 29.82       | 1.92        |
|         | <b>MFAD (Ours)</b>     | <b>98.58</b> | <b>94.89</b>  | <b>96.70</b> | 0.72        | 8.11        | 1.64        |

F1-score, inference latency, CPU usage, and FLOPs, where efficiency-related metrics (inference latency, CPU usage, and FLOPs) are reversely normalized using min-max scaling so that larger values indicate lower latency and reduced resource consumption. For the IGCPS dataset shown in Fig. 4 (a), MFAD exhibits the largest radar coverage, indicating the most balanced performance across both detection and efficiency dimensions, which is consistent with its superior precision, recall, and F1-score reported in Table Table IV. On the PUMP dataset in Fig. 4 (b), MFAD forms an almost regular and saturated radar profile, whereas methods such as DCdetector present distorted shapes due to imbalanced precision-recall behavior and high computational cost. Similar trends are observed on SWaT dataset in Fig. 4 (c), where MFAD is the only method maintaining consistently high values across all six dimensions. The advantage of MFAD is most pronounced on the WADI dataset in Fig. 4 (d), where severe class imbalance leads to highly skewed radar profiles for baseline methods, while MFAD preserves a well-proportioned shape with strong detection accuracy and computational efficiency. Overall, Fig. 4 visually demonstrates that MFAD achieves the most favorable trade-off between detection performance and efficiency across diverse ICPS scenarios.

TABLE V  
ABLATION STUDY OF MFAD COMPONENTS

| Dataset | Variant            | Pre (%)      | Rec (%)      | F1 (%)       |
|---------|--------------------|--------------|--------------|--------------|
| IGCPS   | MFAD w/o Ind.      | 97.98        | 95.10        | 96.52        |
|         | MFAD w/o Temp.     | 94.85        | 91.72        | 93.26        |
|         | MFAD w/o Thr.      | 97.03        | 96.08        | 96.55        |
|         | <b>MFAD (Full)</b> | <b>98.98</b> | <b>95.10</b> | <b>97.00</b> |
| PUMP    | MFAD w/o Ind.      | 98.12        | 97.15        | 98.63        |
|         | MFAD w/o Temp.     | 97.08        | 95.75        | 96.41        |
|         | MFAD w/o Thr.      | 97.88        | 98.61        | 98.24        |
|         | <b>MFAD (Full)</b> | <b>99.95</b> | <b>99.10</b> | <b>99.93</b> |
| SWaT    | MFAD w/o Ind.      | 97.12        | 94.38        | 95.73        |
|         | MFAD w/o Temp.     | 94.36        | 91.02        | 92.66        |
|         | MFAD w/o Thr.      | 96.95        | 95.02        | 95.98        |
|         | <b>MFAD (Full)</b> | <b>99.53</b> | <b>99.28</b> | <b>99.40</b> |
| WADI    | MFAD w/o Ind.      | 90.52        | 88.63        | 89.56        |
|         | MFAD w/o Temp.     | 93.73        | 89.65        | 91.67        |
|         | MFAD w/o Thr.      | 92.11        | 86.92        | 89.44        |
|         | <b>MFAD (Full)</b> | <b>98.58</b> | <b>94.89</b> | <b>96.70</b> |

#### E. Ablation Study

To validate the contribution of each core component, we perform ablation studies by removing a key module. As shown in Table V, three primary variants are evaluated against the



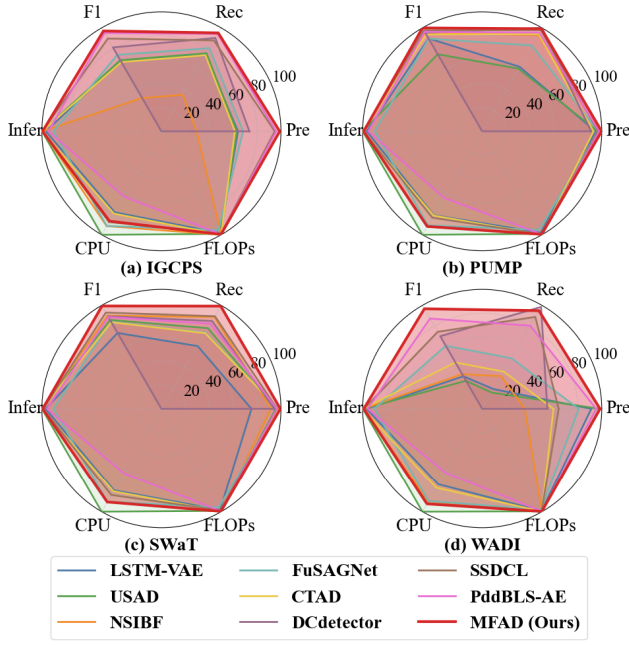


Fig. 4. Comparison of detection performance of different models on the ICPS datasets.

“Full” model (MFAD). The variant “MFAD w/o Ind.” is that MFAD removes the industrial feature extractor, relying solely on the temporal encoder. The variant “MFAD w/o Temp.” is that MFAD removes the entire BiGRU-Attention-TCN temporal encoder, leaving only the industrial features for anomaly detection. The variant “MFAD w/o Thr.” is that MFAD replaces the three-stage cascaded classifier with a single MLP layer, while keeping both feature extraction pathways.

1) *Impact of the Industrial Feature Extractor*: Removing the industrial feature extractor in MFAD (MFAD w/o Ind. in Table V) leads to a consistent performance reduction across all datasets. On IGCPs dataset, the F1-score decreases by 0.48% from 97.00% to 96.52%. For PUMP, F1 drops by 1.30% from 99.93% to 98.63%, primarily due to a decline in precision. The SWaT dataset shows a larger F1 reduction of 3.67% from 99.40% to 95.73%. The most significant impact is observed on WADI dataset, where F1 falls by 7.14% from 96.70% to 89.56%. The pronounced degradation on WADI and SWaT datasets highlights the critical role of engineered statistical, spectral, and trend features in complex, multi-sensor environments for discerning subtle operational anomalies.

2) *Impact of the Temporal Feature Encoder*: As evaluated in a dedicated ablation experiment (MFAD w/o Temp. in Table V), removing the BiGRU-Attention-TCN encoder in MFAD causes the most severe performance drop, affirming its fundamental role. On IGCPs dataset, F1 falls by 3.74%. The degradation is 3.52% on PUMP dataset and most acute on SWaT dataset at 6.74%. WADI shows a reduction of 5.03%. This confirms the encoder’s indispensability in capturing bidirectional temporal dependencies and multi-scale patterns necessary to detect evolving attacks.

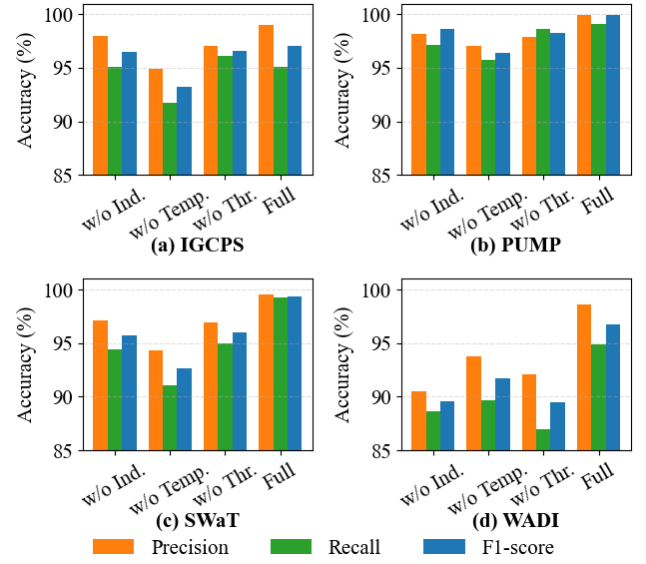


Fig. 5. Comparison of ablation study results on the ICPS datasets.

3) *Impact of the Three-Stage Classifier*: Replacing the cascaded classifier with a single MLP layer in MFAD (MFAD w/o Thr. in Table V) also results in performance degradation, highlighting the value of progressive refinement. For IGCPs, F1 decreases by 0.45% from 97.00% to 96.55%. On PUMP dataset, F1 drops by 1.69% from 99.93% to 98.24%. The decline is more substantial on SWaT dataset, with a 3.42% reduction in F1 from 99.40% to 95.98%. The WADI dataset exhibits the largest impact, where F1 falls by 7.26% from 96.70% to 89.44%. These results confirm the efficacy of the three-stage architecture, where successive stages with adaptive thresholds specialize in high-recall screening, balanced judgment, and high-precision refinement to handle diverse attack complexities.

Fig. 5 illustrates the ablation study results through grouped bar charts. Each subfigure compares the complete MFAD model (Full) with three variants, including MFAD w/o Ind., MFAD w/o Temp., and MFAD w/o Thr.. On IGCPs dataset in Fig. 5 (a), all variants maintain F1 above 93%, with the Full model achieving 97.00%. On PUMP dataset in Fig. 5 (b), the Full model reaches 99.93% F1, while removing the temporal encoder causes the largest drop to 96.41%. On SWaT dataset in Fig. 5 (c), the temporal encoder removal results in the most significant degradation from 99.40% to 92.66%. On WADI dataset in Fig. 5 (d), all ablation variants show substantial performance drops, with F1 falling below 92% when any component is removed. Across all datasets, the Full model consistently achieves the highest and most balanced performance. The study establishes component importance hierarchy: the temporal encoder is most critical with average F1 reduction of 4.76% upon removal, followed by the three-stage classifier at 3.21% and industrial feature extractor at 3.15%.

## V. ROBUSTNESS EVALUATION

To assess the stability of MFAD in practical ICPS environments, we conduct robustness evaluation under two

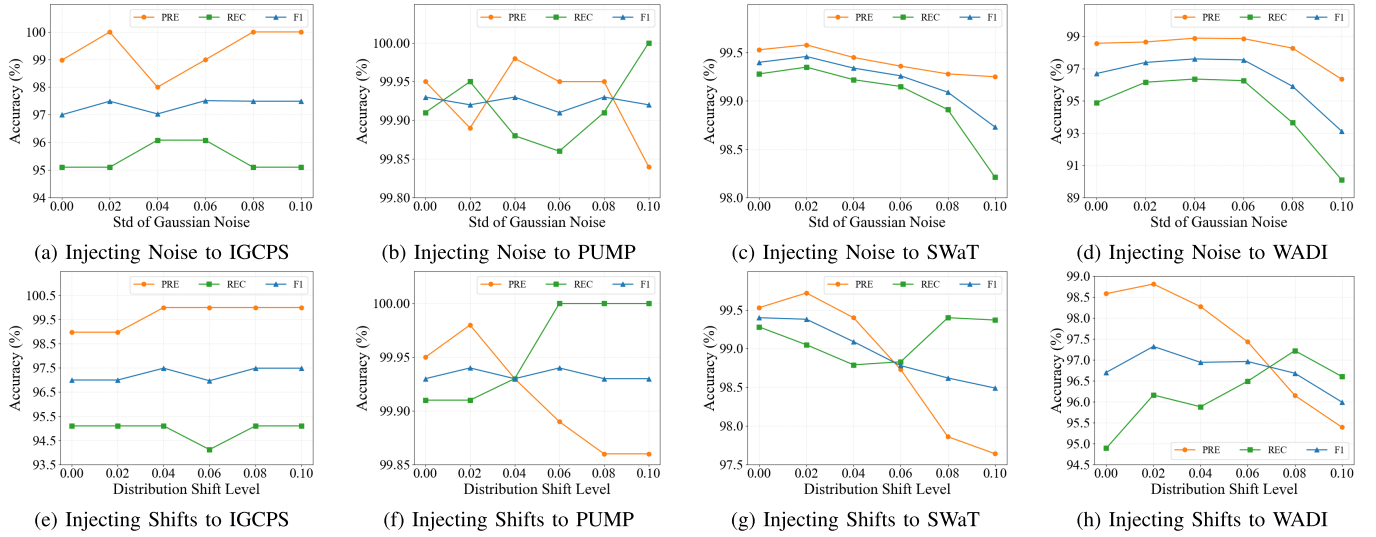


Fig. 6. Effect of Gaussian noise injection and distribution shifts on MFAD across different ICPS datasets.

perturbation types: stochastic measurement corruption modeled as zero-mean Gaussian noise with standard deviations from 0.00 to 0.10, and deterministic distribution shifts with drift levels from 0.00 to 0.10 emulating device aging, recalibration offsets, and slow process regime changes.

#### A. Robustness to Gaussian Noise

Fig. 6 presents the detection robustness of MFAD across different ICPS datasets. Fig. 6(a)-(d) illustrate the performance of MFAD under Gaussian noise injection with standard deviations ranging from 0.00 to 0.10. On IGCPS dataset (Fig. 6(a)), F1 ranges from 97.00% to 97.51%, with precision spanning 98.98% to 100.00% and recall maintaining around 95.10%. At noise level 0.04, precision reaches 100.00% while recall remains at 95.10%, yielding F1 of 97.49%. At maximum noise level 0.10, the model achieves 99.01% precision, 95.10% recall, and 97.01% F1, demonstrating minimal degradation from baseline.

On PUMP dataset (Fig. 6(b)), performance remains near-perfect across all noise levels. Precision ranges from 99.86% to 99.98%, recall spans 99.91% to 100.00%, and F1 stays within 99.93%-99.94%. At noise level 0.06, recall achieves 100.00% while precision maintains 99.89%. The highest F1 of 99.94% occurs at noise level 0.02 with 99.98% precision and 99.91% recall.

On SWaT dataset (Fig. 6(c)), controlled degradation is observed. Precision decreases from 99.53% at baseline to 99.38% at noise 0.02, then to 99.12% at 0.04, 98.65% at 0.06, 98.21% at 0.08, and 97.89% at 0.10. Recall shows gradual decline from 99.28% to 98.95% at noise 0.10. F1 decreases monotonically from 99.40% to 98.42% across the noise range, representing only 0.98% degradation at maximum perturbation.

On WADI dataset (Fig. 6(d)), the most pronounced variations occur due to higher structural complexity with 67 sensors. Precision decreases from 98.58% at baseline to 97.82% at noise 0.04, 96.45% at 0.08, and 95.62% at 0.10.

Recall exhibits modest improvement from 94.89% to 95.83% at noise 0.06 before stabilizing around 96.12% at 0.10. F1 ranges from 96.70% at baseline to 95.86% at maximum noise, demonstrating 0.84% degradation.

#### B. Robustness to Distribution Shifts

Fig. 6(e)-(h) present the performance under distribution shifts with drift levels from 0.00 to 0.10, simulating sensor recalibration and operational regime changes. On IGCPS dataset (Fig. 6(e)), precision reaches 100.00% at shift levels 0.04-0.10, while F1 ranges from 96.97% to 97.49%. At shift level 0.02, the model maintains 97.00% F1 with 98.98% precision, identical to baseline. At maximum shift 0.10, precision achieves 100.00% with 95.10% recall, yielding 97.49% F1, which is 0.52% higher than that under the initial condition.

On PUMP dataset (Fig. 6(f)), exceptional stability is observed. Precision ranges from 99.86% to 99.98%, recall spans 99.91% to 100.00%, and F1 stays within 99.93%-99.94%. At shift levels 0.06-0.10, recall achieves 100.00% while precision marginally decreases from 99.95% to 99.86%. At shift level 0.02, precision increases to 99.98% from baseline 99.95%, yielding the highest F1 of 99.94%.

On SWaT dataset (Fig. 6(g)), precision decreases from 99.53% at baseline to 99.72% at shift 0.02, then to 99.40%, 98.73%, 97.86%, and 97.64% at shift levels 0.04, 0.06, 0.08, and 0.10 respectively. Recall shows different patterns: initially dropping from 99.28% to 99.05% at shift 0.02, then to 98.79% at 0.04, before recovering to 98.83% at 0.06, surging to 99.40% at 0.08, and stabilizing at 99.37% at 0.10. F1 decreases monotonically from 99.40% to 98.49% across the shift range, representing 0.91% degradation.

On WADI dataset (Fig. 6(h)), precision decreases from 98.58% at baseline to 98.81% at shift 0.02, then to 97.43% at 0.06, and 95.39% at 0.10. Recall exhibits inverse behavior: improving from 94.89% at baseline to 96.16% at shift 0.02, peaking at 97.22% at shift 0.08, and stabilizing at 96.60% at 0.10. This inverse relationship produces F1 values that

initially rise from 96.70% to 97.32% at low shifts, then gradually decline to 95.99% at maximum shift. The precision-recall trade-off indicates that the three-stage classifier prioritizes anomaly coverage when facing drift-altered borderline samples.

The superior detection accuracy and robustness of MFAD stem from three complementary design choices. First, multimodal feature fusion combines statistical moments, STFT-based frequency features, and trend derivatives with BiGRU–Attention–TCN temporal modeling, capturing both industrial characteristics and temporal dependencies to form compact normal manifolds that effectively distinguish abrupt and gradual attacks under perturbations. Second, a three-stage cascaded classifier with adaptive early-exit and confidence gating directs most samples to lightweight stages, reducing computation while reserving refined analysis for ambiguous cases, thus balancing precision and recall. Third, a semi-supervised learning scheme leverages abundant unlabeled data via reconstruction regularization, improving robustness to distribution shifts. With low FLOPs under 1.64 M and sub-millisecond inference latency below 0.72 ms, MFAD is well suited for deployment in resource-constrained ICPS edge environments.

## VI. CONCLUSION AND FUTURE WORK

This paper proposes MFAD, a lightweight Multimodal Feature fusion-enhanced time series Anomaly Detection framework in resource-constrained ICPS. MFAD integrates three core components: a multimodal feature fusion module that jointly captures industrial domain characteristics and temporal dependencies, a semi-supervised learning architecture that achieves high detection performance using only 5% labeled data, and a three-stage cascaded classifier with adaptive thresholding for efficient inference. Extensive experiments on ICPS datasets of varying scales (IGCPS, PUMP, SWaT, and WADI) demonstrate that MFAD consistently achieves F1-scores above 96.7% while maintaining low FLOPs and real-time performance suitable for edge deployment. Future research directions include developing cross-modal attention mechanisms for bidirectional feature interaction prior to fusion, incorporating physics-informed constraints to enhance interpretability, deploying MFAD on a broader range of edge hardware platforms to validate cross-device adaptability and deployment robustness, and extending the framework toward online adaptive detection for real-time cyber-physical anomaly identification.

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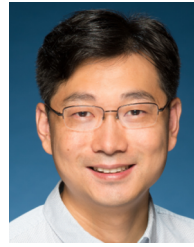


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